

Exam Guide

Automatic Image Analysis ■ learning from test-exam mistakes

Every mistake turned into a rule, a procedure, and a picture

How to use this guide

Each topic follows the same pattern: the **question**, **what went wrong**, the **correct answer**, **why** it is correct, a **step-by-step** procedure that works on any variant of the task, and a short **rule to remember**. Question numbers refer to `test_exam.md`; the same task types appear in the 2021 and 2022 exams.

Error map

Q	Topic	Your answer	Correct	Sec.
2	Classical pipeline order	wrong order	see figure	2
10	Backprop memory	c	a	3
11	Backprop gate rules	a, d	a	3
12	Bias, validation	a, b	a, b, d	4
14	Hough point test	c, d	a	5
16	R-table transform	b ✓	b	6
19	More conv filters	b, c	c, d	7
20	Max pooling	a, c, d	a, b, c	7
21	Conv parameters	252	3024	7
23	Histograms, entropy	1.5 / 2	see table	8
24	Kadir saliency	c	b	8
27	Backprop through time	–	1	9
29	Poisson MLE	–	4.40	10
30	Bayes posterior	–	0.5027	11
31	Discriminant functions	b, c	c, d	12
33	GAN statements	last: true	last: false	13

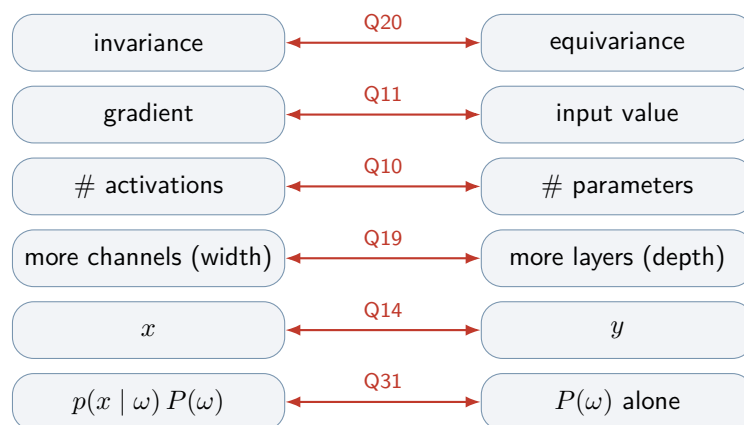
1 General Procedure for Multiple Choice

all questions

Most lost points came from multi-select questions: one option too many (Q11, Q19, Q20, Q31) or one too few (Q12). The knowledge was often there; the checking procedure was not.

Step by step

1. **Judge every option on its own.** Cover the others and ask: “Is this sentence true by itself?” Never stop after the first true option.
2. **Look for absolute words.** “always”, “only”, “necessarily”, “entirely” make a statement false in almost every case.
3. **Check for swapped partner concepts** (figure below). The trap is a true sentence with one word exchanged for its partner.
4. **Check that the option answers the question.** “Higher abstraction” is a true property of CNNs, but not an effect of *more filters in one layer*.
5. **Numeric answers: formula first, numbers second**, then check the order of magnitude.



Partner concepts that were swapped in the test exam. When an option contains one of them, re-read it with the partner and decide which one makes the sentence true.

2 Classical Computer Vision Pipeline

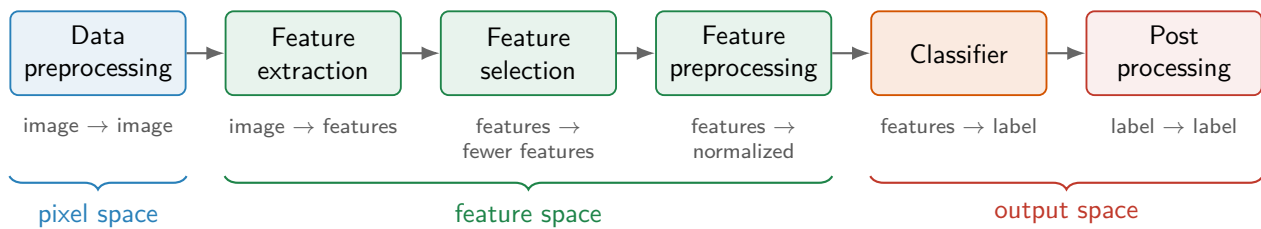
Q2

Question

Bring Data Preprocessing, Feature Selection, Feature Extraction, Feature Preprocessing, Post processing, and Classifier into the correct order.

✓ Correct answer

Data Preprocessing → Feature Extraction → Feature Selection → Feature Preprocessing → Classifier → Post processing.



The six stages, sorted by what goes in and what comes out. Everything in pixel space comes first; everything that needs labels comes last.

Step by step

1. Write down input and output type of each stage (image, features, label).
2. Chain them so that each output type is the next input type.
3. Inside feature space: you can only *select* features that have been *extracted*; you normalize only the features that survive the selection, directly before the classifier.

Remember

Data → Extract → Select → Prepare → Classify → Post.

3 Backpropagation

Q10, Q11

Memory consumption (Q10)

✗ Your answer

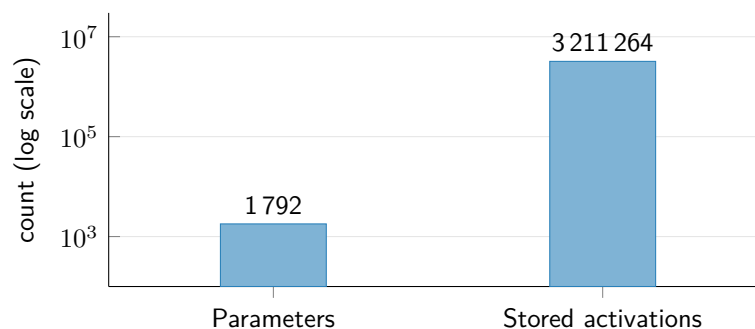
“Memory is proportional to the number of trainable parameters.”

✓ Correct answer

“Memory is proportional to the combined output dimensionalities of the nodes in the network.”

Why

The backward pass needs, at every node, the value that node produced in the forward pass: the input x for the derivative of $w \cdot x$, the ReLU input to know whether it was active. All these **activations** are stored until the backward pass. Their number is the sum of the output sizes of all nodes (times the batch size). Convolutional layers have few parameters but huge outputs, so parameter count and memory are not proportional.



One 3×3 convolution, $3 \rightarrow 64$ channels, on a 224×224 image: under two thousand parameters, but over three million output values that must be kept for the backward pass (per image).

Also: one update step needs exactly **one** forward and **one** backward pass. Options with “multiple passes” are false.

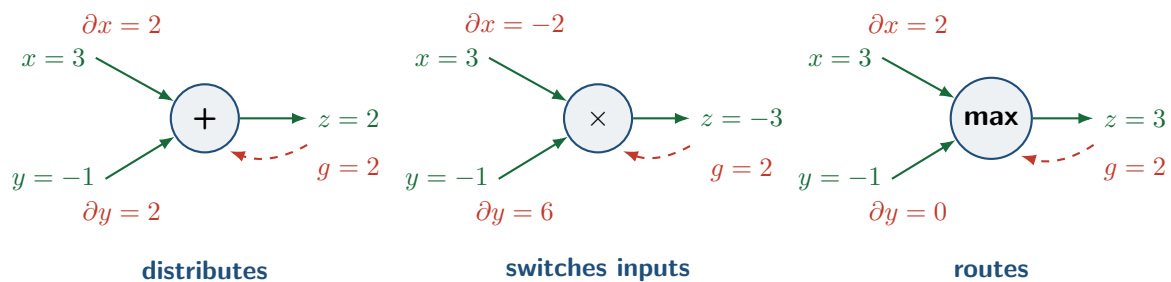
Gate rules (Q11)

✗ Your answer

Ticked “Addition nodes backpropagate their inputs”.

✓ Correct answer

Only “Backpropagation allows for efficient optimization of computational graphs with gradient descent”.



Forward values (green) and backward gradients (red) for the same inputs $x = 3$, $y = -1$ and upstream gradient $g = 2$. Only the multiplication node uses the input values, and it swaps them:

$$\partial x = g \cdot y, \partial y = g \cdot x.$$

Node	Forward	Backward (upstream gradient g)
Addition $z = x + y$	sum	$\partial x = g, \partial y = g$ (copy the gradient)
Multiplication $z = xy$	product	$\partial x = g y, \partial y = g x$ (swap the inputs)
Max $z = \max(x, y)$	maximum	g to the larger input, 0 to the other

Why

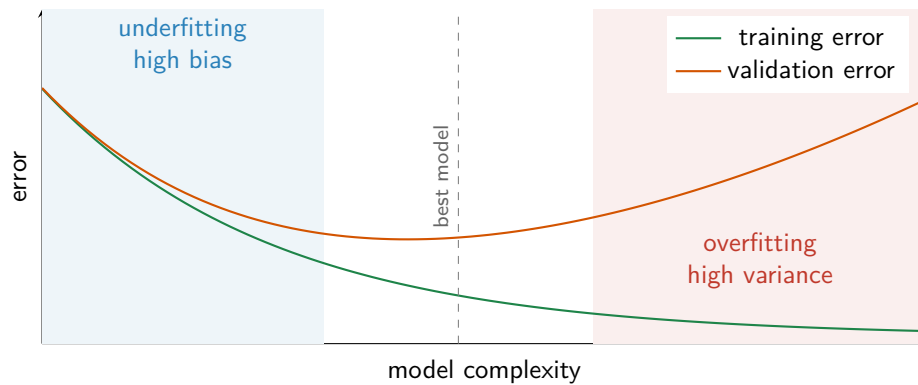
The addition node passes on the *gradient*, never its inputs. Backpropagation also does *not* reduce SGD noise (that comes from mini-batches) and does *not* guarantee the optimum (the loss is non-convex). It only computes the exact gradient efficiently.

4 Bias, Variance, and Model Complexity

Q12

✗ Your answer

a, b (missed d: “A model with high bias tends to underfit the data.”)



Training error always falls with complexity. Validation error falls, then rises again. High bias sits on the left (both errors high), high variance on the right (small training, large validation error).

Statement	True?	Reason
Small training error $\Rightarrow$ sufficient complexity	✓	the model is flexible enough to fit the data
Small validation error $\Rightarrow$ good performance	✓	unseen data estimates generalization
Small training error $\Rightarrow$ good performance	✗	the model could overfit (right side of the plot)
High bias $\Rightarrow$ underfitting	✓	too simple, it misses even the training data (left side)

Remember

Bias = **B**elow the data (underfit). **V**ariance = **V**olatile, fits the noise (overfit).

5 Hough Transform: Point on a Line

Q14

Question

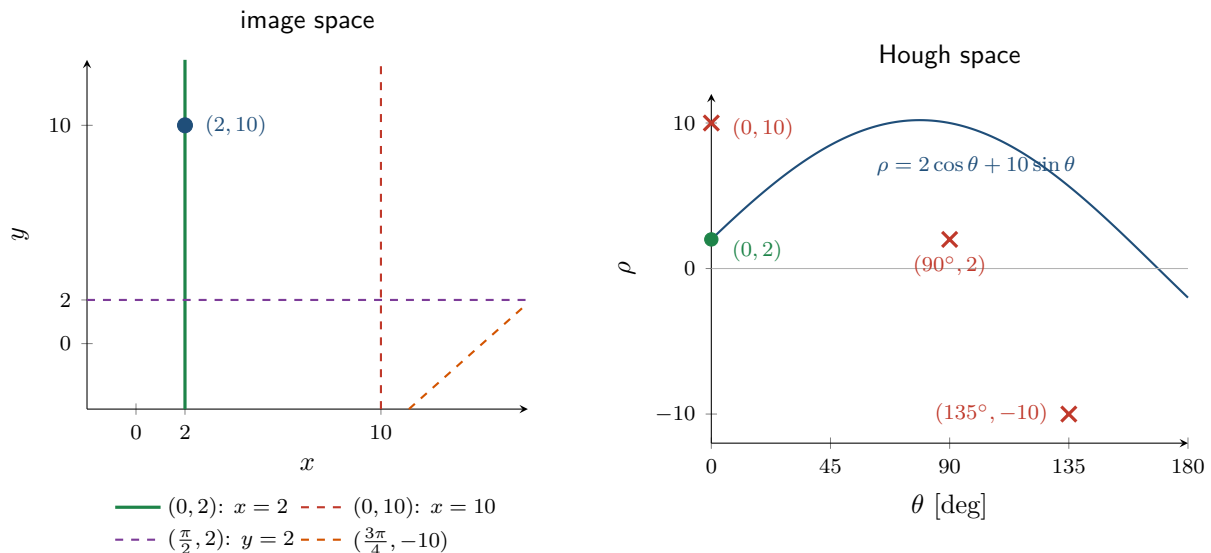
Which lines (θ, ρ) pass through $(x, y) = (2, 10)$? Options: $(0, 2)$, $(0.75\pi, -10)$, $(0, 10)$, $(0.5\pi, 2)$.

✗ Your answer

$(0, 10)$ and $(0.5\pi, 2)$ — the roles of x and y were swapped.

✓ Correct answer

Only $(\theta, \rho) = (0, 2)$.



Step by step

1. Write the line equation: $\rho = x \cos \theta + y \sin \theta$.
2. Insert the point: $\rho(\theta) = 2 \cos \theta + 10 \sin \theta$.
3. Evaluate every option and compare with the given ρ :

Option	$\cos \theta$	$\sin \theta$	$2 \cos \theta + 10 \sin \theta$	given ρ	
(0, 2)	1	0	2	2	✓
$(0.75\pi, -10)$	-0.707	0.707	5.66	-10	✗
(0, 10)	1	0	2	10	✗
$(0.5\pi, 2)$	0	1	10	2	✗

Remember

$\theta = 0$: $\rho = x$, the line is **vertical** ($x = \rho$). $\theta = \pi/2$: $\rho = y$, the line is **horizontal** ($y = \rho$). Always compute; never match numbers by eye.

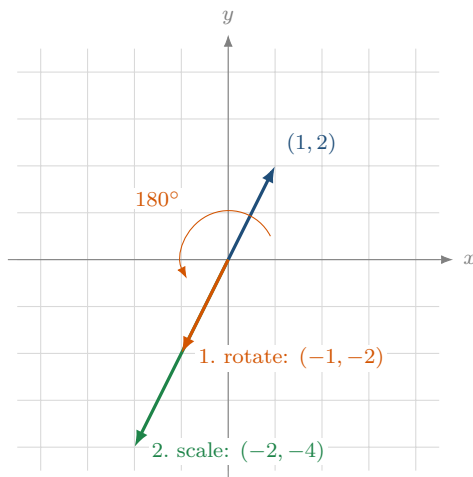
6 Generalized Hough Transform: R-Table

Q16

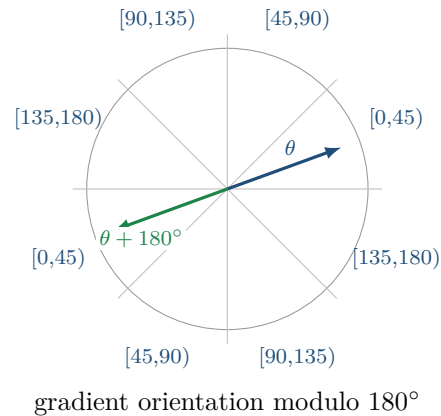
Answered correctly (b). The reasoning below keeps it reproducible for other scales, angles, and table ranges.

Question

Transform the R-table with scale $s = 2$ and a counter-clockwise rotation of $\varphi = 180^\circ$. Rows: $[0, 45)$: (1, 2); (3, 4) $[45, 90)$: – $[90, 135)$: (5, 6); (7, 8) $[135, 180)$: (9, 10).



displacement vectors

gradient orientation modulo 180°

Left: rotating by 180° negates each displacement, scaling by 2 doubles it. Right: a gradient direction rotated by 180° lies on the same line, so with a table over $[0^\circ, 180^\circ)$ it stays in its row.

Step by step

1. **Scale:** multiply every displacement by s . Rows stay, since scaling does not change gradient directions.
2. **Rotate the vectors** by φ . For 180° : $(d_x, d_y) \mapsto (-d_x, -d_y)$.
3. **Rotate the rows:** every entry moves to the row for $\theta + \varphi$. Check the table range! Over $[0^\circ, 180^\circ)$, orientations are taken modulo 180° , so $\theta + 180^\circ \equiv \theta$: nothing moves.
4. Result: $(1, 2) \rightarrow (-2, -4)$, $(3, 4) \rightarrow (-6, -8)$, $(5, 6) \rightarrow (-10, -12)$, $(7, 8) \rightarrow (-14, -16)$, $(9, 10) \rightarrow (-18, -20)$; rows unchanged. $\Rightarrow$ **b**.

Remember

Distractors: only shifted rows (a), swapped components = reflection (c), unchanged (d). If the table covers $[0^\circ, 360^\circ)$, a 180° rotation *does* move rows. Translation never changes an R-table.

7 CNN Layers

Q19, Q20, Q21

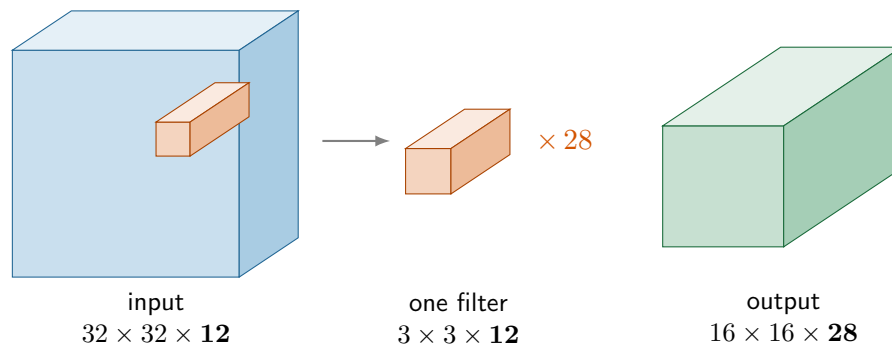
Number of weights (Q21)

✗ Your answer

$252 = 3 \cdot 3 \cdot 28$ — the input channels were forgotten.

✓ Correct answer

$3 \cdot 3 \cdot 12 \cdot 28 = 3024$



Every filter spans the *full input depth* ($3 \times 3 \times 12 = 108$ weights). One filter produces one output channel, so 28 filters are needed. The spatial sizes ($32 \rightarrow 16$) do not enter the count; they only reveal the stride.

Step by step

$$\# \text{weights} = K_h \cdot K_w \cdot C_{\text{in}} \cdot C_{\text{out}} = 3 \cdot 3 \cdot 12 \cdot 28 = 3024, \quad \# \text{biases} = C_{\text{out}} = 28.$$

Stride (Q22): $W_{\text{out}} = \left\lfloor \frac{W_{\text{in}} + 2P - K}{S} \right\rfloor + 1$. Halving the size ($148 \rightarrow 74$ with $K = 3$, $P = 1$) means $S = 2$.

More filters in one layer (Q19)

✗ Your answer

Ticked “Higher abstraction capabilities of the layer”.

✓ Correct answer

Increase in output dimensionality; increased number of features the layer can respond to.

Change	What it adds	Abstraction?
More filters (width)	more output channels = more feature detectors at the same level	✗
More layers (depth)	larger receptive field, features of features	✓

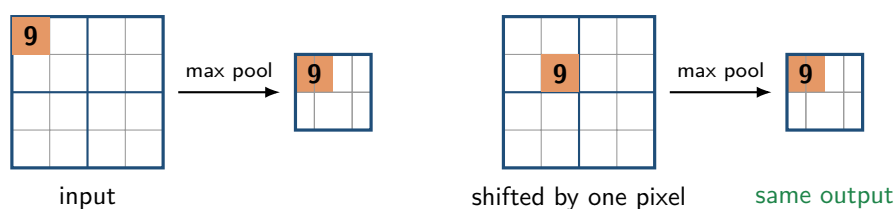
Effects of max pooling (Q20)

✗ Your answer

Ticked “Translation equivariance” instead of “Translation invariance”.

✓ Correct answer

Dimensionality reduction, translation invariance, loss of spatial/relational information.



A small shift inside a 2×2 pooling block does not change the pooled output: **invariance**. The price is that the exact position is lost. An *equivariant* operation (convolution) would shift its output along with the input.

Remember

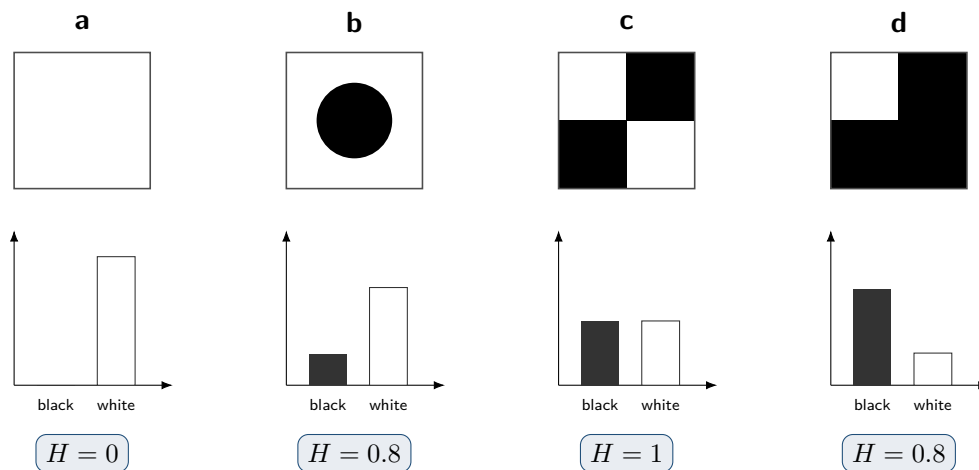
Conv = Equivariant (output moves with the input). **Pool = Invariant** (output stays).

8 Histograms, Entropy, and Saliency

Q23, Q24

Matching histograms and entropy values (Q23)

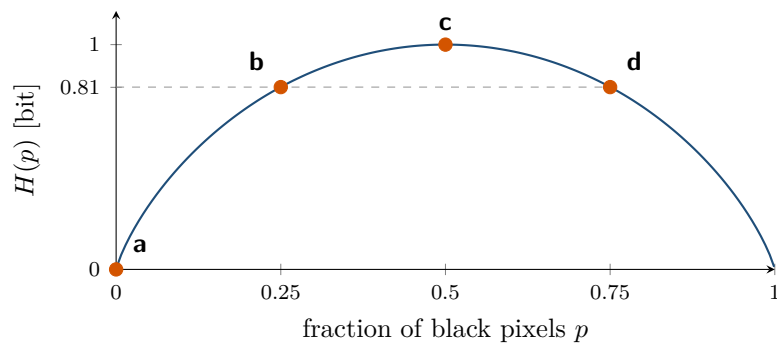
The task had 8 drag-and-drop placements (4 histograms, then 4 entropy values), each worth 0.25 points. A score of 1.5/2 means two wrong placements, typically **one swapped pair**.



Correct matching. The histogram counts only *how many* black and white pixels there are, never *where* they are. Patches b and d have mirrored histograms but the same entropy.

Step by step

1. **Fix the axis orientation first.** The single-bar histogram must belong to the all-white patch a. Its bar sits on the right, so the axis runs from black (left) to white (right).
2. For each patch, estimate the black/white fraction and pick the histogram with those bar heights. Mostly white (b): tall bar right. Mostly black (d): tall bar left.
3. Compute $H = -\sum_i p_i \log_2 p_i$: one value $\Rightarrow 0$; $50/50 \Rightarrow 1$; $25/75 \Rightarrow 0.81$.
4. The two “0.8” tokens look identical but may be graded as distinct items. Place each next to the histogram it belongs to.



Binary entropy is symmetric: 25 % black and 75 % black give the same 0.81 bit.

Most salient patch after Kadir et al. (Q24)

✗ Your answer

c (the patch with the highest entropy).

✓ Correct answer

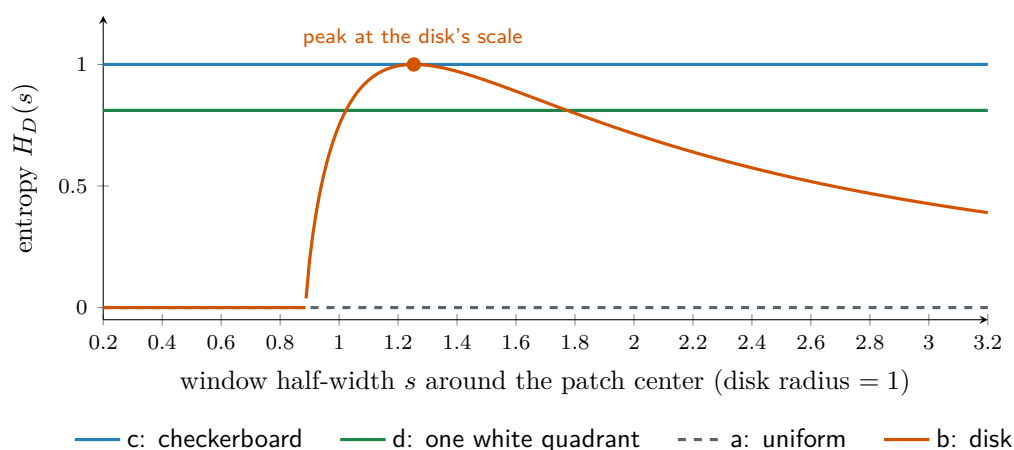
b (the black disk).

Why

Kadir–Brady saliency is **not** entropy alone:

$$y(s_p, x) = H_D(s_p, x) \cdot W_D(s_p, x), \quad W_D(s, x) = s \int \left| \frac{\partial}{\partial s} p(I, s, x) \right| dI.$$

H_D is taken at a scale s_p where the entropy has a **peak** over scale, and W_D is large only if the histogram **changes across scale**. A patch whose statistics look the same at every window size scores zero, however high its entropy.



Entropy as a function of window size, window centered on each patch. Checkerboard, quadrant, and uniform patch keep the same black/white ratio at every size: flat curves, no peak, $W_D = 0$. Only the disk shows a peak where the window just encloses it; there the histogram changes fastest.

Step by step

1. Grow a window around the center of each patch and imagine its histogram.
2. Does the histogram change with window size? If not, the entropy curve is flat: no peak, not salient (a, c, d).
3. Is there a window size where the entropy is maximal and then falls again? That is a *characteristic scale*: salient (b).

Remember

Kadir saliency favors **isolated blobs with a characteristic scale**, not the maximum-entropy patch. Entropy alone is insufficient because the intensity distribution contains **no spatial information** (Q25).

9 Backpropagation Through Time

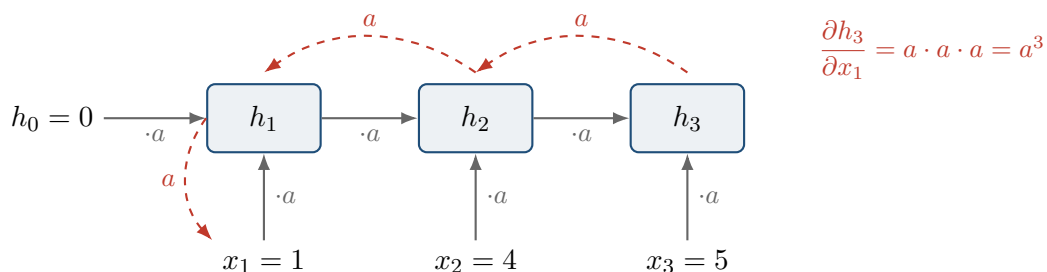
Q27

Question

RNN cell $h_t = a h_{t-1} + a x_t$ with $a = 1$, $h_0 = 0$, sequence $(x_1, x_2, x_3) = (1, 4, 5)$. What is the derivative with respect to x_1 ?

✓ Correct answer

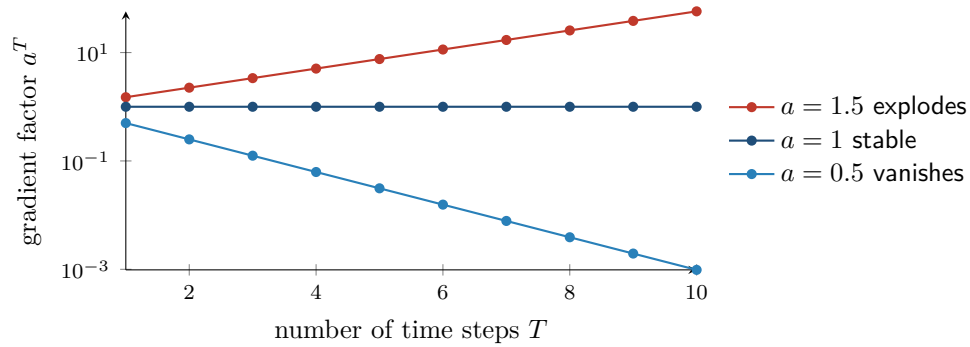
$$\partial h_3 / \partial x_1 = a^3 = 1$$



Unrolled over three time steps with shared weight a . The gradient travels backwards (red) and picks up one factor a per step.

Step by step

1. **Unroll:** $h_1 = ax_1$, $h_2 = a^2x_1 + ax_2$, $h_3 = a^3x_1 + a^2x_2 + ax_3$.
2. **Chain rule backwards through time:** $\frac{\partial h_3}{\partial x_1} = \frac{\partial h_3}{\partial h_2} \cdot \frac{\partial h_2}{\partial h_1} \cdot \frac{\partial h_1}{\partial x_1} = a^3$.
3. **Insert $a = 1$:** result 1. The values $x_2 = 4$, $x_3 = 5$ are distractors, because the cell is linear.



The same product a^T explains the vanishing and exploding gradient problem of RNNs.

10 Maximum Likelihood Estimation

Q29

Question

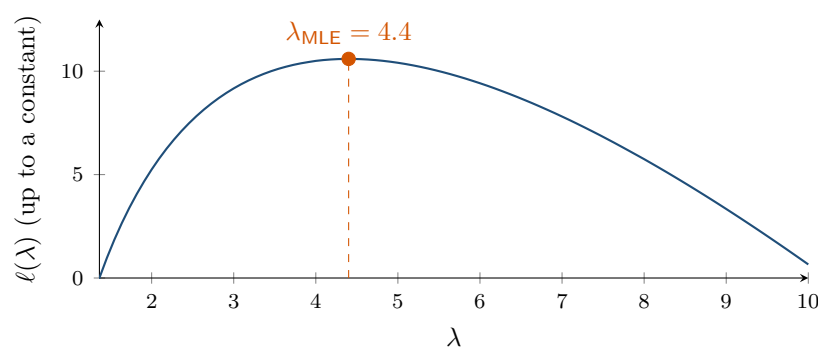
Error counts $X = \{2, 8, 4, 2, 6\}$, model $P(k) = \lambda^k e^{-\lambda} / k!$. Find λ_{MLE} .

✓ Correct answer

$$\lambda_{\text{MLE}} = \bar{k} = 22/5 = 4.40$$

Step by step

1. Likelihood (product over samples): $L(\lambda) = \prod_i \lambda^{k_i} e^{-\lambda} / k_i!$
2. Log-likelihood: $\ell(\lambda) = (\sum_i k_i) \ln \lambda - n\lambda - \sum_i \ln k_i!$
3. Derivative = 0: $\frac{\sum_i k_i}{\lambda} - n = 0 \Rightarrow \lambda_{\text{MLE}} = \frac{1}{n} \sum_i k_i$
4. Insert: $(2 + 8 + 4 + 2 + 6)/5 = 22/5 = 4.40$



The log-likelihood $22 \ln \lambda - 5\lambda$ peaks at the sample mean.

Distribution	MLE
Poisson λ	sample mean
Bernoulli p	fraction of ones
Gaussian μ	sample mean
Gaussian σ^2	$\frac{1}{n} \sum_i (x_i - \hat{\mu})^2$ (divide by n , not $n - 1$)
Exponential λ	$1/\text{sample mean}$

11 Bayesian Posterior

Q30

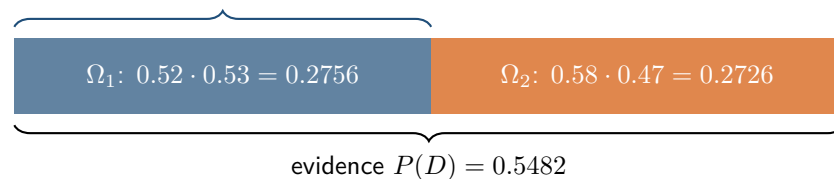
Question

$P(D | \Omega_1) = 0.52$, $P(D | \Omega_2) = 0.58$, $P(\Omega_1) = 0.53$, $P(\Omega_2) = 0.47$. Find $P(\Omega_1 | D)$ to four decimals.

✓ Correct answer

0.5027

$$P(\Omega_1 | D) = 0.2756 / 0.5482 = 0.5027$$



Each bar is likelihood $\times$ prior. The posterior is one bar's share of the total. Ω_2 has the higher likelihood, but its lower prior makes Ω_1 slightly more probable.

Step by step

1. Numerator per hypothesis: likelihood $\times$ prior.
2. Evidence: sum of *all* numerators.
3. Posterior: own numerator / evidence. Check that all posteriors sum to 1 ($0.5027 + 0.4973$).
4. Keep four or more decimals in between; round only at the end.

12 Discriminant Functions

Q31

✗ Your answer

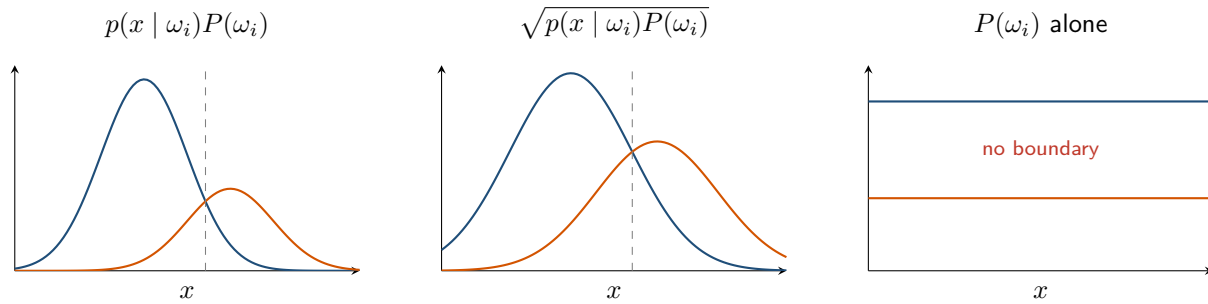
$P(\omega_i)$ and $\sqrt{p(x | \omega_i)P(\omega_i)}$ — missed $P(\omega_i | x)$.

✓ Correct answer

$\sqrt{p(x | \omega_i)P(\omega_i)}$ and $P(\omega_i | x)$.

Why

$g_i(x)$ is a valid discriminant function if it **ranks the classes like the posterior** $P(\omega_i | x)$ for **every** x . The posterior itself qualifies trivially, and so does any strictly increasing transform applied equally to all classes. $P(\omega_i)$ ignores x : it picks the same class for every input.



Two classes with priors 0.7 and 0.3. The square root changes the curves but not where they cross, so the decision boundary (dashed) is the same. The priors alone never cross: every x goes to class 1.

Candidate $g_i(x)$	Valid?
$P(\omega_i x)$	✓ always (the reference)
$p(x \omega_i)P(\omega_i)$	✓ always (posterior $\times p(x)$, same for all classes)
$\sqrt{\cdot}, \ln(\cdot)$ of the above	✓ always (strictly increasing)
$p(x \omega_i)$	✗ only if all priors are equal
$P(\omega_i)$	✗ only if all likelihoods are identical
$1/g_i, -g_i$	✗ never (ranking reversed)

Remember

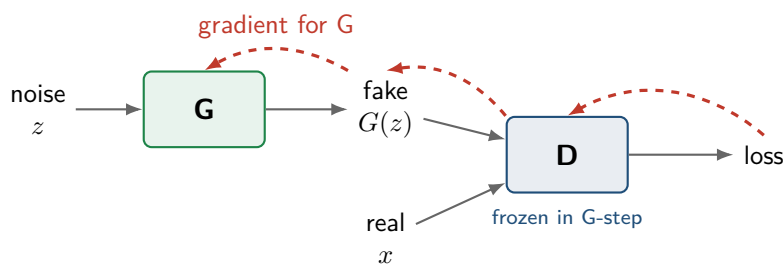
Quick test: does it use **likelihood and prior together** (or the posterior directly), combined by an increasing function? Then it is valid.

13 Generative Adversarial Networks

Q33

✗ Your answer

Marked “The computation of backpropagation for the generator is independent of the discriminator” as true.



The generator’s loss is measured at the discriminator’s output. Its gradient must travel back *through* D (with D’s weights frozen) to reach G. Without D there is no training signal for G.

Statement	Answer	Reason
Generator learns the (empirical) probability function	✗ False	G is an <i>implicit</i> model: a sampler $z \mapsto x$, not a density you can evaluate
All GANs require labeled data (x, y)	✗ False	the basic GAN needs only real images; real/fake labels are generated automatically
CycleGAN trains two generators and two discriminators	✓ True	$G: X \rightarrow Y, F: Y \rightarrow X$, plus D_X, D_Y and the cycle loss
Generator backprop is independent of the discriminator	✗ False	the gradient flows through D (figure above)

Remember

Explicit density estimators: MLE, Parzen windows, (restricted) Boltzmann machines. A GAN only *samples*. Check whether the question asks for “estimating a distribution” or “generating samples” (Q28).

14 Checklist Before Submitting

- ☐ Multi-select: every option judged on its own?
- ☐ “always”, “only”, “necessarily” checked?
- ☐ Invariance/equivariance, inputs/gradients, width/depth not swapped?
- ☐ Hough: $\rho = x \cos \theta + y \sin \theta$ computed for every option?
- ☐ Conv parameters: $K \cdot K \cdot C_{\text{in}} \cdot C_{\text{out}}$, spatial size irrelevant?
- ☐ R-table: orientation range (mod 180° or mod 360°) checked?
- ☐ Histograms: axis orientation fixed with the unambiguous patch?
- ☐ Saliency: entropy *peak over scale*, not maximum entropy?
- ☐ Bayes: evidence = sum over *all* hypotheses, round only at the end?
- ☐ Discriminant: likelihood *and* prior (or the posterior)?
- ☐ BPTT: unroll, one factor per time step, ignore distractor values?